

also become quicker with new software. One example is XENDEE, a San Diego tech company, who have brought out an advanced toolkit for microgrids which makes turnkey solutions available in up to 90 percent lesser time and cost than traditional methods.

India's microgrid market is taking off as well, with the aim to bring power to rural areas using renewable energy. In 2016, the Ministry of New and Renewable Energy in India launched a plan to install 1,000 microgrids and minigrids. One notable microgrids project in India is CREDA's installation of 500 solar microgrids in Chhattisgarh.

### ENERGY BLOCKCHAIN AND IOT

Blockchain technology has found a calling, apart from recording cryptocurrency transactions – the energy market. The decentralised structure of Blockchain makes it ideal for eliminating the middlemen when it comes to supplying electricity. It rids issues pertaining to energy distribution inequality and allows consumers to buy and sell energy with each other directly.

Furthermore, IoT devices connected to the blockchain-enabled energy sources can autonomously buy and sell energy at optimal times, and also optimise energy system settings in real-time and contextually. These solutions can also monitor and scrutinise the performance of energy-consuming devices to let consumers know how efficiently, or inefficiently, they're using energy. Adaptation of this technology is seen in projects such as Brooklyn Microgrid, which is a community-powered microgrid which features a community energy market.

### GRID PARITY AND DECLINING COSTS

Grid parity is achieved when an alternative renewable energy source generates power at a lower or equal cost, and has a higher or equal performance level, than traditional power sources. Solar and wind energy have reached parity in both cost and performance in multiple regions. This tears down the traditional top-down, centrally-distributed energy model and ushers in an evenly-distributed and need-based model.

Solar and wind energy have proven to be efficient, and more affordable than conventional sources. The use of these renewable energy sources drive up economic benefits and have a low environmental impact, making them a win-win.

### NEXT-GEN RENEWABLES

Conversations and policies surrounding renewable and sustainable energy generally fail to induct two next-gen sources – offshore wind power and enhanced geothermal systems – which should make their way into the energy market by 2030, as per predictions. Offshore wind turbines can be built far into the ocean and can be constructed to be much taller than their onshore counterparts. This boosts their operating efficiencies. The US DOE (Department of Energy) stated that over 23,000MW of offshore wind projects could be initiated by 2030 which will generate 3 to 4 percent of the country's total electricity.

EGS or enhanced geothermal system is still in its nascent stages of development but sports massive potential. This process involves drilling deep underground wells, then circulating a certain

fluid over naturally heat-producing rocks, and using the steam generated in a closed-loop system to spin turbines placed on the surface. Also, this technique is not limited by existing geology and terrains. The DOE has accelerated research and expects commercially viable technology to be ready in 2030.

### WHERE INDIA STANDS

India invested over \$90 billion in renewable energy over the last ten years and places sixth on the list of countries with the highest investments in clean energy. India began its solar revolution in the year 2015 and though it was late to the



Pavagada solar park, Karnataka, India, aims 100GW of solar power by 2020

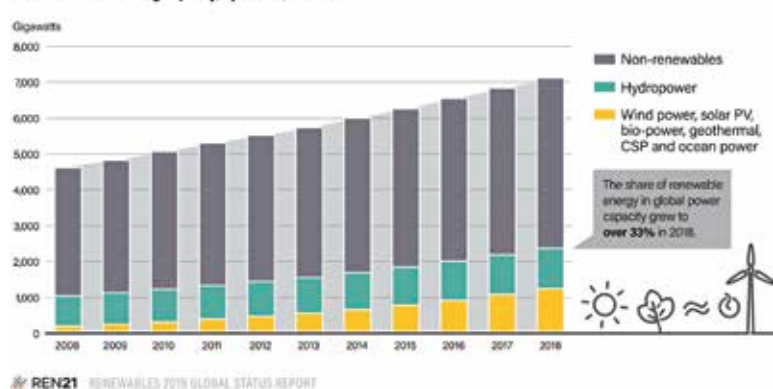
party, it has quickly caught up. We have reached a historic low of ₹2.44 per unit (or kWh) mostly due to the installation of several solar parks.

Surprisingly, India actually lowered investments in solar energy in 2018 when compared to 2017. Reports state that the investments dropped due to the falling capital costs.

Unfortunately, the government's 2019-2020 budget allocated 63 percent higher funds to coal-based power in comparison to alternative fuels. The Ministry of Coal received ₹20,121 crores while the Ministry of New and Renewable Energy was granted ₹12,353 crores.

Despite this, the Central Electrical Authority of India reported that solar and wind energy projects may constitute more than half of India's total power capacity by 2030. PM Narendra Modi had set a climate target of having 40 percent non fossil fuel-powered capacity by 2030 and by the looks of figures, the projections will be exceeded by 2030. The government aimed to install 175GW worth of renewable capacity by 2022 and as of May 2019, we have already reached 80GW. If all goes as planned, we may be headed towards a more sustainable and progressive future. 

Global Power Generating Capacity, by Source, 2008-2018



Global power generating capacity: 2008-2018